

From Genes to Cell Types: Leveraging LLMs for Cell Type Annotation

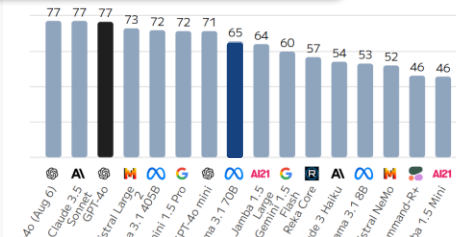


Yuanyuan Wang Ph.D

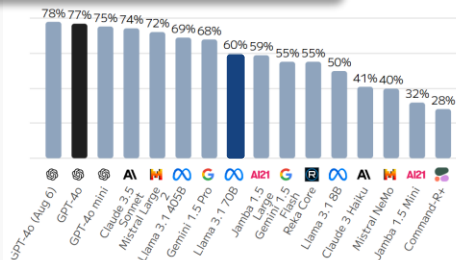
LLMs: Large Language Models

Can GPT4 or Llama3 Understand Genomics Data?

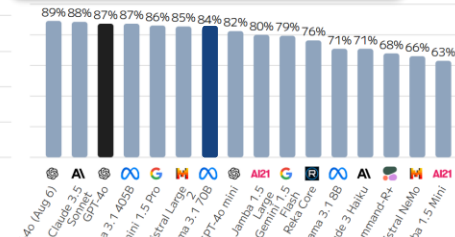
Artificial Analysis Quality



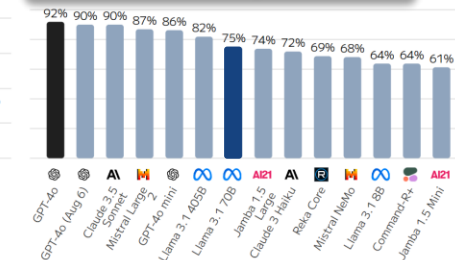
Math



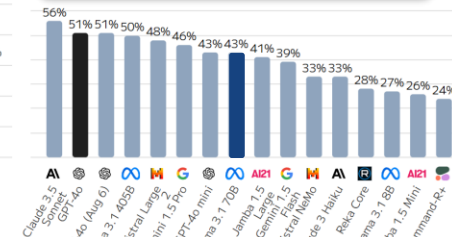
Reasoning & Knowledge



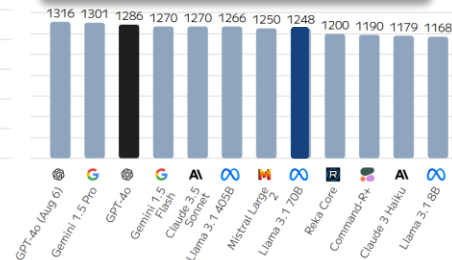
Coding



Scientific Reasoning



Communication



Llama3, an open-source model, offers a more cost-effective alternative to GPT-4.

Improve accessibility in Bioinformatics: No Code Solutions



An example Prompt to LLMs:



- Marker genes are highly expressed in specific cell types, making them essential for cell type annotation in scRNA-seq analysis.
- I have a group of blood cells with the top N differentially expressed genes
- ranked by average fold change compared to other cells:

"LEF1", "TRAT1", "CISH", "LINC02273", "TESPA1", "SIRPG", "PCED1B", "DPP4", "CD40LG",
"TNFRSF25" ...
- **Given these genes,**
- **could you predict the cell type for this group of cells?**



Benchmarking LLMs for Cell Type Prediction

2

Models

Choose the best one or the cost-effective one?

GPT4 vs. Llama3 (Open-sourced, cost-effective)

14

Prompts

Does it matter how we ask the questions?

2

Bioinformatics Tools

Can tools improve marker gene selection?

Scanpy vs. Seurat

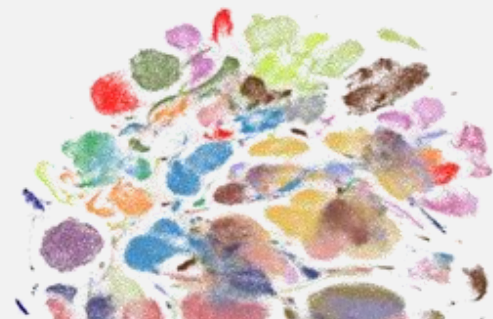
6

Datasets

Does performance vary across tissues?

Benchmark Data: Tabula Sapiens Human Cell Atlas

We Focused on 5 Tissues and Immune Cells



500,000 cells
24 tissues/organs
+400 cell types

- Blood
- Bone Marrow
- Lung
- Spleen
- Thymus
- Immune Cells



81 Unique Cell Types

Consortium Members



Tabula Sapiens Consortium*, Jones RC, Karkanias J, et al.
The Tabula Sapiens: A multiple-organ, single-cell transcriptomic atlas of humans. *Science*. 2022;376(6594):eabl4896. doi:10.1126/science.abl4896

The Power of Prompt:



There is only one "e" in the word "gene."

**Because LLM can only see tokens (numerical),
Not words.**

You

How many e are in the word gene?

The Power of Prompt:



There is only one "e" in the word "gene."

You

How many e are in the word gene?



Certainly! Let's break it down step by step.

The word "gene" has four letters: G, E, N, and E.

To determine how many "e"s are in the word, we can go through each letter and count how many of them are "e."

Starting with the first letter, G, there is no "e" in it.

Moving on to the second letter, E, there is one "e" in it.

Next is the third letter, N, which does not contain an "e."

Finally, we have the fourth letter, E again. This also contains one "e."

Adding up the number of "e"s from each letter gives us a total of two "e"s in the word "gene."

So, to answer your question: There are two "e"s in the word "gene."

You

Let's think step by step.

Prompt converts letter counting to logical reasoning.



Benchmark Setup: 14 Prompt Combinations

Prompt :

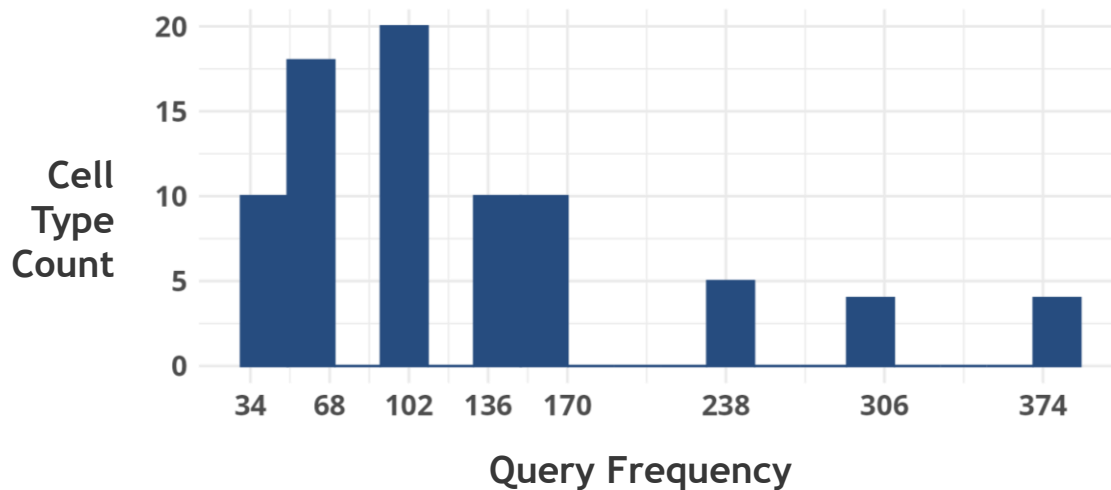
- Marker genes are highly expressed in specific cell types, making them essential for cell type annotation in scRNA-seq analysis. **[Add examples]**
- I have a group of **[blood cells]** with the **[top 100]** differentially expressed genes ranked by average fold change compared to other cells:
 "LEF1", "TRAT1", "CISH", "LINC02273", "TESPA1", "SIRPG", "PCED1B", "DPP4", "CD40LG", "TNFRSF25" ...
 - Given these genes,
 - could you predict the cell type for this group of cells?

Features added	Genes	10					25			50			100	
		1	2	3	4	5	6	7	8	9	10	11	12	13
Example		●	●	●	●	●	●	●	●	●	●	●	●	●
Tissue Type		●	●	●	●	●	●	●	●	●	●	●	●	●
Chain of Thought		●	●	●	●	●	●	●	●	●	●	●	●	●
Prompt Combinations		1	2	3	4	5	6	7	8	9	10	11	12	13



Ensuring Reliability: High Query Frequency Across Cell Types

- For each query, each of the 14 prompts were repeated 5 times.
- Common cell types are often present across multiple tissues.
- **On average, each cell type was queried 102 times.**



Evaluation Metrics: Can LLMs give consistent results?



- When measuring *a single prediction* for a cell type:

Mismatch

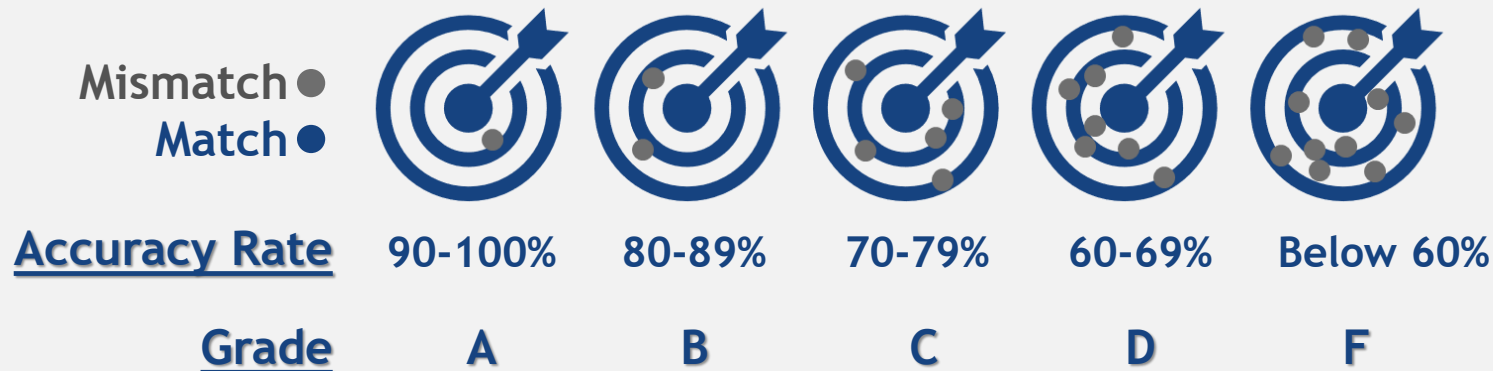
Match

Prediction:
Label (Correct Answer):



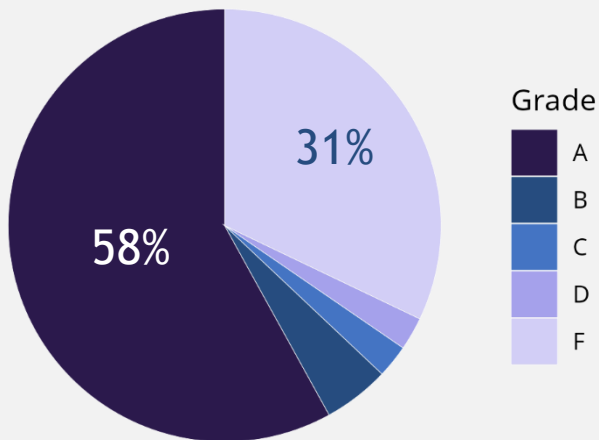
Evaluation Metrics: Can LLMs give consistent results?

- When averaging *multiple* predictions for a cell type:



Cell Type Level Accuracy:

- Using the best prompt:



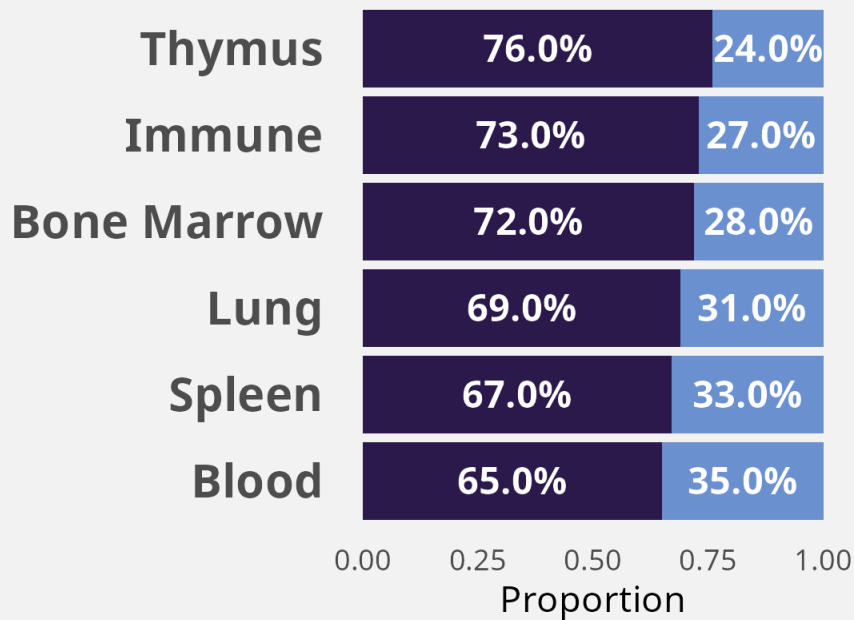
- Even with just 10 genes, GPT can predict:

B cells,
CD4+ helper T cell,
CD4+ alpha-beta T cell,
CD8+, alpha-beta cytotoxic T cell,

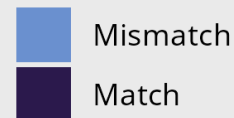
Langerhans cell,
Erythroid lineage cell,
Granulocyte,
Mast cell,
Mature conventional dendritic cell,

Neutrophil,
Regulatory T cell,
Type II Pneumocyte.

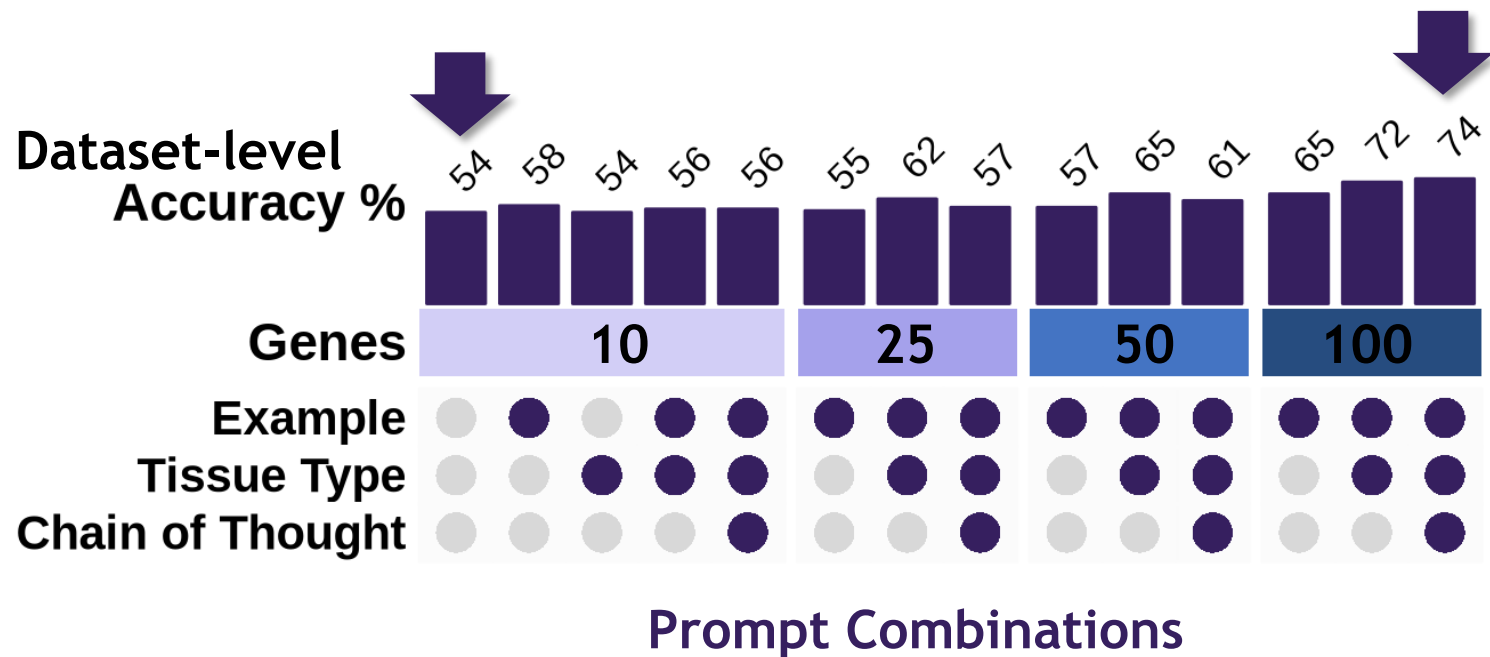
Dataset Level Accuracy: Variation Across Tissues



GPT Prediction vs.
Label (Correct Cell Type)



Boosting Accuracy with Better Prompt Design



Failures Are Just as Important

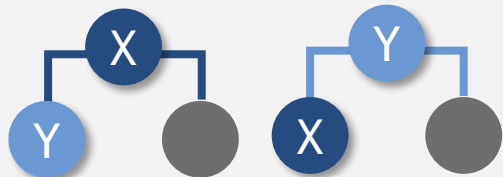
Predicted Cell Type: Y

Correct Cell Type: X

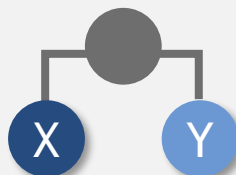


Mismatch can be categorized by the following relationships:

Child-Parent



Siblings



Unrelated



A Graph Approach to evaluate Failures

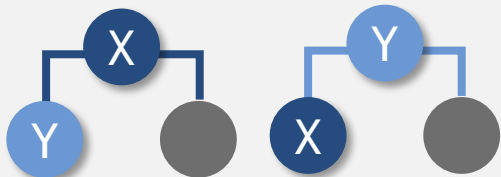
Predicted Cell Type: Y

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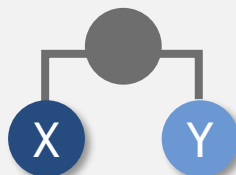


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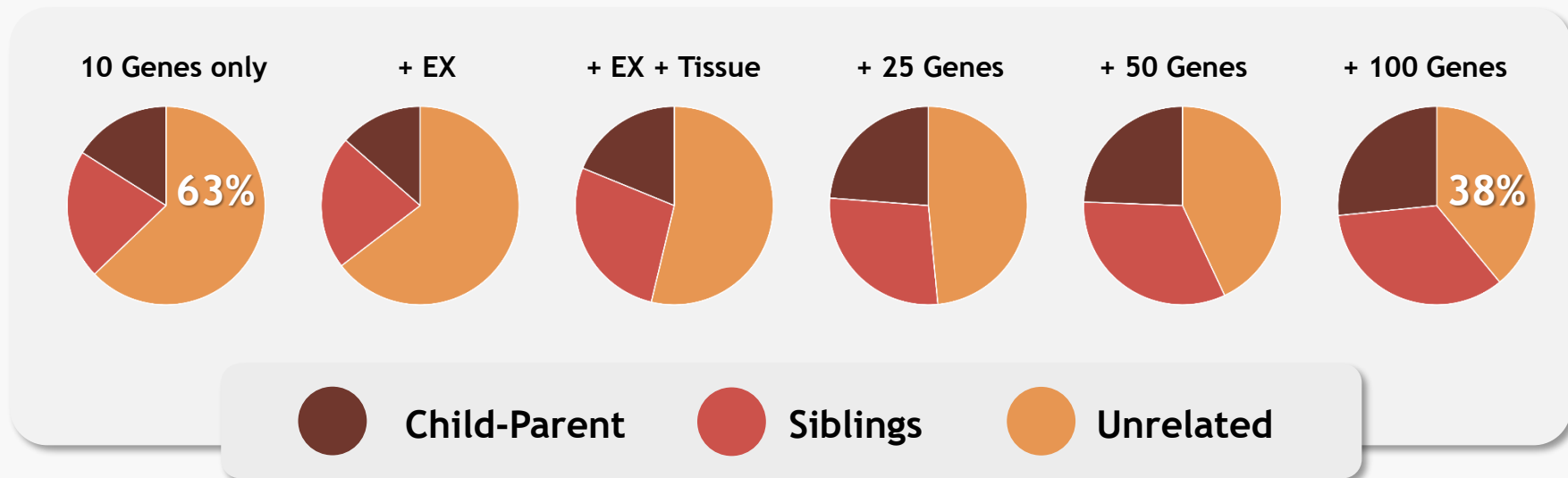
Siblings



Unrelated



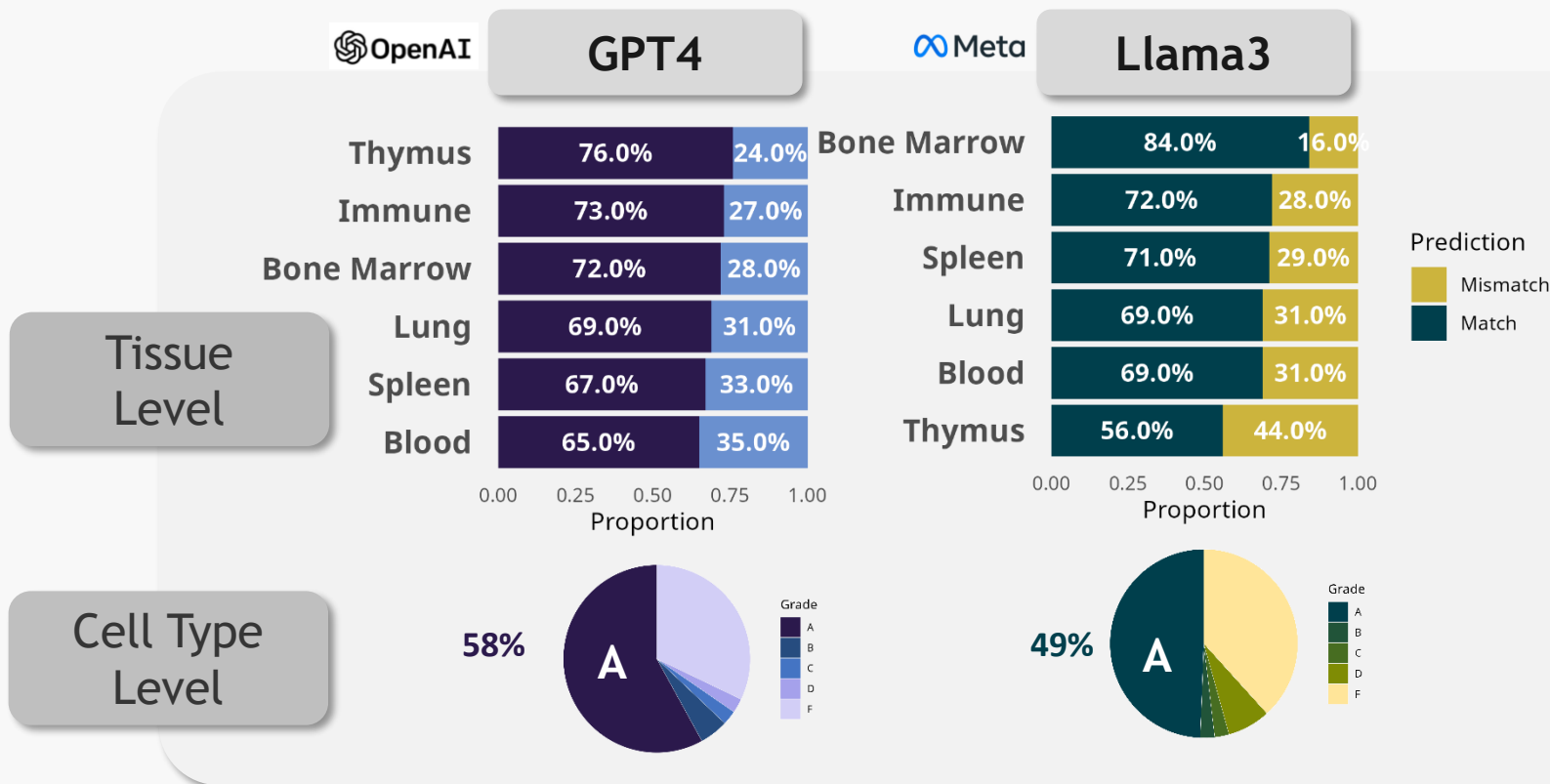
Close Calls: Better Prompts Reduce Unrelated Predictions



With better prompting, unrelated cases reduced from 63% to 38%.

- + EX: add examples
- + Tissue: add tissue types

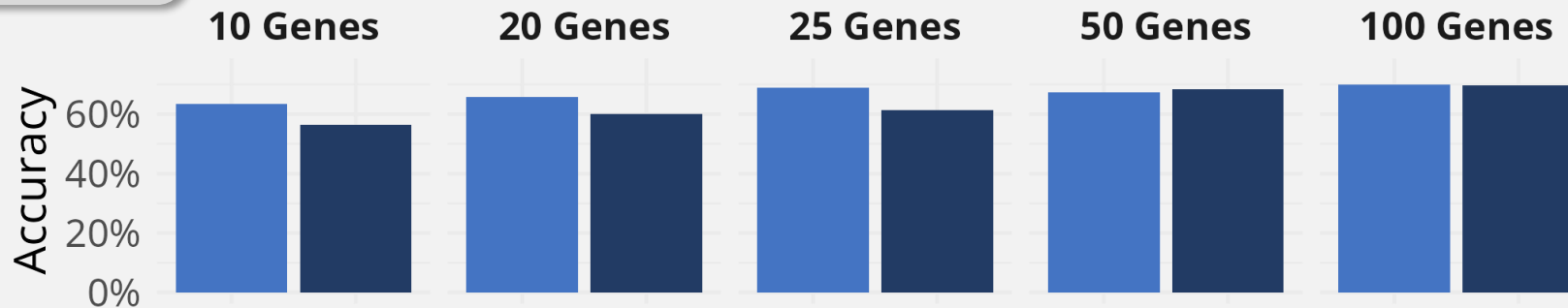
Llama3: The Open-Source Competitor



Bioinformatics Tools: *When Does it Matter*



Method



Key Takeaways



LLMs provide consistent and accurate results with smart prompts.

Accuracy varies by tissue—
Thymus and Immune cells lead the pack

Open-source models show great promise for boosting predictions.